LESSON 5.8

Determine Surface Area of Pyramids

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the surface area of a pyramid by adding the base area and the lateral faces.

WHY THIS MATTERS Your class is building a pyramid-shaped display to showcase the time capsule at the school entrance. The pyramid will be covered in gold leaf to make it shine. To figure out how much gold leaf you need, you must calculate the total surface area — the base plus all the triangular faces! The pyramid has a square base that is 10 inches on each side, and each triangular face has a slant height of 12 inches.

REMEMBER

Determine Surface Area of Pyramids

Surface Area of Pyramids. Formula: Surface Area = Base Area + Sum of Triangular Lateral Faces Surface area of a pyramid is the base area plus all the triangular side faces. A square pyramid has 1 square base and 4 triangular faces. Each triangle uses the slant height: ½ × base × slant height.

1. Calculate area of base polygon (square, rectangle, or triangle).
2. Calculate area of each triangular face using ½ × base × slant height.
3. Add the base area and all triangular face areas together

Words: **Pyramid *(Pirámide)*** **Slant height *(Altura inclinada (apotema))*** **Lateral face *(Cara lateral)*** **Base *(Base)*** **Apex *(Ápice)***

1 **Circle the letter of the best answer. Show how you know.**

How many faces does a triangular prism have?

(A) 3 (B) 4

(C) 5 (D) 6

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A prism's faces have areas 10 cm², 10 cm², 6 cm², 6 cm², 15 cm² and 15 cm². What is the surface area?

(A) 31 cm²
(B) 52 cm²
(C) 62 cm²
(D) 62 cm³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

A triangular prism has two triangular bases each with area 12 cm² and three rectangular faces with areas 30 cm², 40 cm², and 40 cm². What is the total surface area?

(A) 110 cm²
(B) 122 cm²
(C) 134 cm²
(D) 134 cm³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

What is the surface area of a rectangular prism with l = 5 cm, w = 4 cm, h = 3 cm?

(A) 47 cm²
(B) 60 cm²
(C) 94 cm²
(D) 94 cm³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

A museum is building a square pyramid display case. The base edge is 3 feet and the slant height is 4 feet. The glass costs $12 per square foot. What is the area of one triangular face?

(A) 3 ft²
(B) 6 ft²
(C) 7 ft²
(D) 12 ft²

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

6 **Circle the letter of the best answer. Show how you know.**

A museum is building a square pyramid display case. The base edge is 3 feet and the slant height is 4 feet. The glass costs $12 per square foot. What is the total glass area for all four faces?

(A) 12 ft²
(B) 24 ft²
(C) 30 ft²
(D) 33 ft²

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

EXPLAIN YOUR THINKING

Display B has the largest surface area (280 in²). What contributes more to the total — the base or the lateral faces? Why?

SENTENCE STARTER For Display B, the base area is ___ in² and the total lateral area is 4 × ___ = ___ in². The ___ contribute(s) more because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each square-pyramid gift box (base edge × slant height) to its total surface area.

_____	Base 5 in, slant 8 in	(A)	105 in²
_____	Base 6 in, slant 7 in	(B)	120 in²
_____	Base 8 in, slant 9 in	(C)	208 in²
_____	Base 10 in, slant 6 in	(D)	220 in²

2 **Write the name of the correct group on each line.**

A gift box can be a square pyramid (4 triangle faces) or a rectangular pyramid (a rectangle base with 4 triangle faces). Sort each measurement into the box part it describes.

Groups: **Base (bottom)** **Triangular faces**

_____ Square: side × side

_____ Rectangle: length × width

_____ ½ × base edge × slant height

_____ Uses the slant height as its height

_____ Add 4 of these together

_____ Only one of these on the box

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A museum is building a square pyramid display case. The base edge is 3 feet and the slant height is 4 feet. The glass costs $12 per square foot. What will the glass cost?

(A) $72

(B) $144

(C) $288

(D) $396

PER 1

4 **Circle the letter of the best answer. Show how you know.**

Your first gift box is a square pyramid with a base edge of 7 in and a slant height of 9 in. How much wrapping paper (total surface area) covers it?

(A) 126 in²

(B) 144 in²

(C) 175 in²

(D) 175 in³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

5 **Circle the letter of the best answer. Show how you know.**

Before adding the base, you need the area of just one triangular face of a gift box with base edge 8 in and slant height 5 in. What is it?

(A) 13 in²
(B) 20 in²
(C) 20 in³
(D) 40 in²

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

6 **Circle the letter of the best answer. Show how you know.**

A gift box is a square pyramid with a base edge of 9 in. What is the area of its square base (the part that sits on the table)?

(A) 18 in²
(B) 36 in²
(C) 81 in²
(D) 324 in²

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

EXPLAIN YOUR THINKING

Display B has the largest surface area (280 in²). What contributes more to the total — the base or the lateral faces? Why?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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